Grade 11 · Algebra II · Topic 6: Exponential and Logarithmic Functions · Lesson 6-5 — Properties of Logarithms

Choose Your Route

Three routes, same destination. Start anywhere — you may switch routes at any time, and you may try more than one.

Student name :	Date : .....

PRACTICE — Secure the method

Expand it, condense it, evaluate it

Items 1–3: expand each expression using the properties. Items 4–6 are below.

1.	2.	3.
$\log_2(32x)$	$\log_5 \frac{x}{25}$	$\log_7(49x^3)$

4. Condense $3\log_2 x + \log_2 5$ to a single logarithm.
5. Evaluate $\log_4 30$ to 3 decimal places. Write the ratio first.
6. Evaluate $\log_{27} 9$ exactly, without a calculator.

Done when: each property is named beside the step that uses it, and every decimal is rounded only at the last step.

APPLY — Use it in context

Given values, a sequence and one real context

1. Given $\log_b 2 = 0.3$ and $\log_b 3 = 0.48$ (model values), find $\log_b 12$ and $\log_b 1.5$.
2. A geometric sequence has $a_n = 4(1.5)^{n-1}$. Find the first term that is greater than 50.
3. A model for cruise bookings at a Red Sea terminal (invented for practice) says there are 800 bookings in year 1, rising by 25% each year. Write the year n in which bookings first pass $4\,000$ as a logarithm, evaluate it with a calculator, and decide the year.

Done when: the factors of each number are written before any property is used, and the year number is a whole number checked by testing the year before and the year itself.

INVESTIGATE — Find out why

Arguments, not answers

1. Is $\log_2(x+4) = \log_2 x + 2$? Test it with $x = 4$ and say what the correct statement about a product would be.
2. Use the change of base formula to show that $\log_a b \cdot \log_b a = 1$.
3. Evaluate $\log_7 50$ twice, once with LOG and once with LN. Explain why the two ratios are equal.

Done when: each argument starts from the definition or from numbers, not only from the formula.

Remember

These are names for the tasks, not names for you. Move between them whenever you want to. Everyone answers the same “Time to Check” at the end of the lesson.